

[1]V.-D. Hoang, M.-H. Le, D. C. Hernandez, and K.-H. Jo, “Localization estimation based on Extended Kalman filter using multiple sensors,” *IECON 2013 - 39th Annual Conference of the IEEE Industrial Electronics Society*, pp. 5498–5503, Nov. 2013, doi: <https://doi.org/10.1109/iecon.2013.6700032>.

https://www.researchgate.net/publication/259772429_Localization_estimation_based_on_Extended_Kalman_filter_using_multiple_sensors

This article was used to reference the original goal of creating a particle filter. This was considered an option during the brainstorming phase, as it gave credibility to the idea of using a camera-like sensor. It gave a similar example of camera parameters being used for localization and mapping. It discussed how camera visuals and perspective can make localizing an autonomous vehicle difficult.

[2] Farooq, U. (2024). SensorServer (Version V6.0.4)[An open source android app that exposes the sensor suite on an android device]. Available: <https://github.com/UmerCodez/SensorServer.git>. [Accessed: 2025].

This is the GitHub repository that was referenced in an attempt to try and gain new sensor information for the robot. This repository was specifically used to expose the sensors from an android phone with the consideration of using it to find the heading angle. It could then send these to a web socket to be used in scripts or viewed through a web server.

[3]“AprilTag,” *Umich.edu*, 2016. <https://april.eecs.umich.edu/software/apriltag.html> (accessed Nov. 04, 2025).

<https://april.eecs.umich.edu/software/apriltag.html>

This is the webpage of the AprilTag lab. This is the development page for the AprilTag app. The resources from this page led to the development that we tried to use to gain a second camera angle.